

Saving and Loading Files

Making data outlive the program

THE PROBLEM

Everything so far disappears

Everything we've built lives only in the program's **memory** - it disappears the moment the program ends.

File I/O is how a program remembers something between separate runs.

WRITING

Writing a file

`std::ofstream` ("output file stream") opens a file for writing.

```
#include <fstream>

// The file is created in the working directory.
// Where the program is run from.
std::ofstream outputFile("friends.txt");

for (const std::string& friendName : friends) {
    outputFile << friendName << "\n";
}

outputFile.close();
```

Write to it with `<<`, just like `std::cout`.

READING

Reading a file back

`std::ifstream` ("input file stream") opens a file for reading.

```
std::ifstream inputFile("friends.txt");
std::vector<std::string> friendsFromDisk;

std::string line;
while (std::getline(inputFile, line)) {
    friendsFromDisk.push_back(line);
}

inputFile.close();
```

Same `std::getline` already used for keyboard input.

THE KEY IDEA

It's all just a stream

A file and the keyboard are both just a **stream of text** as far as C++ is concerned.

DIRECTION	TOOL
keyboard in	<code>std::cin</code>
console out	<code>std::cout</code> / <code>std::print</code>
file in	<code>std::ifstream</code>
file out	<code>std::ofstream</code>

DEBUG A FILE WRITE

Watch the file appear

Step over the `saveFriendsToFile` -equivalent write, then open `friends.txt` in your IDE's file browser - it now exists, with the friend names written to it.